

Path Tracking Control with State Estimation based on Constrained Zonotopes for Aerial Load Transportation*

Brenner S. Rego¹, Davide M. Raimondo², Guilherme V. Raffo^{1,3}

Abstract—This work deals with the problem of path tracking of a suspended load using a tilt-rotor UAV. Such task requires the knowledge of the load position, which may not be provided by available sensors. Therefore, to accomplish the path tracking, a set-valued state estimator based on constrained zonotopes is proposed to solve the problem of estimating the load position and orientation, considering sensors with different sampling times, and unknown-but-bounded disturbances. Moreover, to provide feedback to the controller based on the estimated state set, an optimal state choice is given according to a proposed constrained minimum-variance criteria. The path tracking is then solved by a discrete-time mixed $\mathcal{H}_2/\mathcal{H}_\infty$ controller with pole-placement constraints. Results from numerical experiments, performed in a platform based on the Gazebo simulator with a Computer Aided Design (CAD) model of the system, are presented to corroborate the performance of the set-valued state estimator along with the designed controller.

I. INTRODUCTION

The load transportation problem arises in a variety of tasks, for instance transportation of containers in harbors [1] and aerial delivery of supplies in search-and-rescue missions [2], among others. An important issue is the recurrent necessity of knowing the load position to accomplish the task. Since available sensors are often embedded in the mobile platform, information on the load position may not be directly available.

The problem of estimating the load position is commonly addressed through visual systems and Bayesian state estimators [3]. However, such algorithms require knowledge on statistical properties of existing process and measurement disturbances, which may not be easily obtained. On the other hand, set-valued estimation requires knowledge only on bounds of existing disturbances. These techniques are based on the construction of sets that include, with guarantee (if bounds are exact), the system states consistent with available measurements [4].

Unmanned aerial vehicles (UAVs) have been extensively used for load transportation, thanks to their versatility and

autonomous operation [5]. Control objectives include path tracking of the UAV with load swing attenuation [6], and path tracking of the suspended load [5]. The latter is the appropriate goal for tasks requiring precise maneuvering of the load, for which the knowledge of the load position is of utmost importance.

The use of tilt-rotor UAVs for load transportation is recent [7]. They are convertible aircrafts capable of performing vertical take-off and landing, hovering flights, and also flights with improved forward speed, offering many advantages over rotary-wing and fixed-wing configurations, providing an enlarged flight envelope in comparison with quadrotor UAVs, for instance. The problem of path tracking of the suspended load has been addressed in [7], [8], in which the position of the load is estimated by an unscented Kalman filter, and a set-valued estimator based on zonotopes, respectively. However, the former method is not guaranteed, while the latter is considerably conservative, and requires iterative procedures to take into account multiple measurements.

This work solves the path tracking problem of a suspended load using a tilt-rotor UAV, through set-valued state estimation based on the recently introduced *constrained zonotopes* [9]. Such sets are an extension of zonotopes, and correspond to an alternative representation for convex polytopes. Unlike zonotopes, several operations in set-valued estimation methods can be computed directly and exactly using constrained zonotopes with low computational burden, resulting in reduced conservatism. The realistic scenario in which the load position is not measured is taken into account, considering sensors with different sampling times, and unknown-but-bounded disturbances. In order to provide feedback to the controller based on the estimated state set, which is a constrained zonotope, a novel optimal state choice is proposed taking into consideration a *constrained minimum-variance criteria*, resulting in the main contribution of this work. Path tracking control for the proposed aerial load transportation task is then designed by a robust discrete-time mixed $\mathcal{H}_2/\mathcal{H}_\infty$ state-feedback approach. The estimation and control strategies are validated through numerical experiments, performed in a platform based on the Gazebo simulator with a Computer Aided Design (CAD) model of the system, to corroborate the performance of the proposed method. This paper improves the previous work [8] by: (i) the design of a far more accurate set-valued state estimator; (ii) a novel criteria for improved feedback connection based on constrained minimum-variance; (iii) improved control design taking into account the time-varying nature of the trajectory; and (iv) rigorous validation of the proposed strategies con-

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sidering the CAD model of the system.

II. TILT-ROTOR UAV WITH SUSPENDED LOAD MODELING

A. Equations of motion

This work regards the tilt-rotor UAV with suspended load (Fig. 1) as a multi-body mechanical system composed of: (i) the aircraft's main structure (including batteries, landing skids, electronics and instrumentation); (ii) the right thruster group (right thruster and revolute joint); (iii) the left thruster group (left thruster and revolute joint); and (iv) the load group (load and rope). For control purposes, the rope is assumed to be rigid and massless.

The equations of motion of the entire system are formulated considering the load as a free rigid body, with the tilt-rotor UAV rigidly attached to it. This choice of perspective allows state-feedback control strategies to directly steer the trajectory of the load [7]. Nine reference frames are defined (Fig. 1): (i) the inertial reference frame, $\mathcal{I}$; (ii) the load center of mass frame, $\mathcal{L}$; (iii) the aircraft's geometric center frame, $\mathcal{B}$; (iv) the main body center of mass frame, $\mathcal{C}_1$; (v) the right thruster center of mass frame, $\mathcal{C}_2$; and (vi) the left thruster center of mass frame, $\mathcal{C}_3$; (vii) the rope's point of connection frame, $\mathcal{A}_1$; (viii) the right tilting axis frame, $\mathcal{A}_2$; and (ix) the left tilting axis frame, $\mathcal{A}_3$.

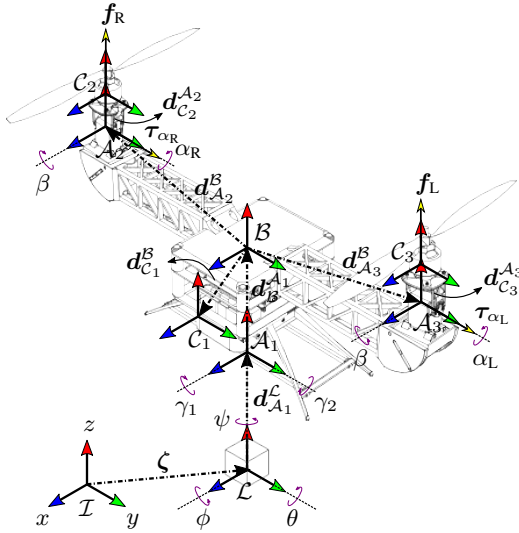


Fig. 1. The tilt-rotor UAV with suspended load, kinematic definitions, input forces and torques.

The position of the load with respect to $\mathcal{I}$ is denoted by $\zeta \triangleq [x \ y \ z]^T$. The displacement from $\mathcal{L}$ to $\mathcal{A}_1$ is defined in $\mathcal{L}$ as $d_{\mathcal{A}_1\mathcal{L}}^{\mathcal{L}} \triangleq [0 \ 0 \ l]^T$, where l corresponds to the rope's length. The displacements from $\mathcal{A}_1$ to $\mathcal{B}$, from $\mathcal{B}$ to $\mathcal{C}_1$, from $\mathcal{B}$ to $\mathcal{A}_i$, and from $\mathcal{A}_i$ to $\mathcal{C}_i$ are denoted by $d_{\mathcal{B}\mathcal{A}_1}^{\mathcal{B}}$, $d_{\mathcal{C}_1\mathcal{B}}^{\mathcal{B}}$, $d_{\mathcal{A}_i\mathcal{B}}^{\mathcal{B}}$, $d_{\mathcal{C}_i\mathcal{A}_i}^{\mathcal{B}}$, respectively, in the respective preceding frames, with $i \in \{2, 3\}$. The orientation of the load with respect to $\mathcal{I}$ is parametrized by Euler angles, $\eta \triangleq [\phi \ \theta \ \psi]^T$, using the ZYX convention about local axes, and is described by

$$R_{\mathcal{L}}^{\mathcal{I}} \triangleq R_{z,\psi} R_{y,\theta} R_{x,\phi} = \begin{bmatrix} c_\psi c_\theta & c_\psi s_\theta s_\phi - s_\psi c_\phi & c_\psi s_\theta c_\phi + s_\psi s_\phi \\ s_\psi c_\theta & s_\psi s_\theta s_\phi + c_\psi c_\phi & s_\psi s_\theta c_\phi - c_\psi s_\phi \\ -s_\theta & c_\theta s_\phi & c_\theta c_\phi \end{bmatrix}.$$

with $s_{(\cdot)} \triangleq \sin(\cdot)$ and $c_{(\cdot)} \triangleq \cos(\cdot)$. The orientation of the UAV w.r.t. $\mathcal{L}$ is described by $\gamma \triangleq [\gamma_1 \ \gamma_2]^T$, with $R_{\mathcal{A}_1}^{\mathcal{L}} \triangleq R_{x,-\gamma_1} R_{y,-\gamma_2}$. The orientation of the thrusters' groups w.r.t. $\mathcal{B}$ are defined by $R_{\mathcal{A}_2}^{\mathcal{B}} \triangleq R_{x,-\beta} R_{y,\alpha_R}$ and $R_{\mathcal{A}_3}^{\mathcal{B}} \triangleq R_{x,\beta} R_{y,\alpha_L}$, where α_R and α_L are the right and left tilting angles, respectively, and β is a fixed inclination angle of the thrusters towards $\mathcal{B}$ [8]. Moreover, $\mathcal{A}_1$, $\mathcal{B}$, and $\mathcal{C}_1$ are parallel and attached to the same rigid body, thus $R_{\mathcal{B}}^{\mathcal{A}_1} = R_{\mathcal{C}_1}^{\mathcal{B}} = R_{\mathcal{C}_1}^{\mathcal{A}_1} = \mathbb{I}_3^1$, $R_{\mathcal{B}}^{\mathcal{L}} = R_{\mathcal{A}_1}^{\mathcal{L}}$, $R_{\mathcal{C}_i}^{\mathcal{B}} = R_{\mathcal{A}_i}^{\mathcal{B}}$, $i \in \{2, 3\}$.

Let $\omega_{\mathcal{L}}^{\mathcal{L}}$ denote the angular velocity of frame $\mathcal{L}$ with respect to $\mathcal{I}$, expressed in $\mathcal{L}$. Then, $\omega_{\mathcal{L}}^{\mathcal{L}} = W_\eta \dot{\eta}$, $\omega_{\mathcal{L}\mathcal{A}_1}^{\mathcal{A}_1} = Q \dot{\gamma}$, $\omega_{\mathcal{B}\mathcal{A}_2}^{\mathcal{A}_2} = a_y \dot{\alpha}_R$, $\omega_{\mathcal{B}\mathcal{A}_3}^{\mathcal{A}_3} = a_y \dot{\alpha}_L$, $\omega_{\mathcal{A}_1\mathcal{B}}^{\mathcal{B}} = \omega_{\mathcal{B}\mathcal{C}_1}^{\mathcal{C}_1} = \omega_{\mathcal{A}_i\mathcal{C}_i}^{\mathcal{C}_i} = \mathbf{0}_{3 \times 1}$, $\omega_{\mathcal{L}\mathcal{B}}^{\mathcal{B}} = \omega_{\mathcal{L}\mathcal{A}_1}^{\mathcal{A}_1}$, and $\omega_{\mathcal{B}\mathcal{C}_i}^{\mathcal{C}_i} = \omega_{\mathcal{B}\mathcal{A}_i}^{\mathcal{A}_i}$, with

$$W_\eta \triangleq \begin{bmatrix} 1 & 0 & -s_\theta \\ 0 & c_\phi & s_\phi c_\theta \\ 0 & -s_\phi & c_\phi c_\theta \end{bmatrix}, \quad Q \triangleq \begin{bmatrix} -c_{\gamma_2} & 0 \\ 0 & -1 \\ s_{\gamma_2} & 0 \end{bmatrix}, \quad a_y \triangleq \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}.$$

The generalized coordinates are defined according to $q \triangleq [\zeta^T \ \eta^T \ \gamma^T \ \alpha_R \ \alpha_L]^T$, which include the load's position and orientation given in the inertial frame and reflect the choice of perspective. Therefore, through the Euler-Lagrange formulation [8], the equations of motion are written as

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + g(q) = \vartheta(q, \dot{q}, u, d), \quad (1)$$

where $M(q)$ is the inertia matrix, $C(q, \dot{q})$ is the Coriolis and centripetal forces matrix, $g(q)$ is the gravitational forces vector, and $\vartheta(q, \dot{q}, u, d) = L_{\text{in}}(q)u - L_{\text{fr}}\dot{q} + L_{\text{db}}d$, with $u = [f_R \ f_L \ \tau_{\alpha_R} \ \tau_{\alpha_L}]^T \in \mathbb{R}^4$ denoting the system inputs (thrust forces and servomotor torques, see Fig. 1), and $d \in \mathbb{R}^3$ denoting external disturbances affecting the load (see [10] for details).

Finally, by defining the system states $x \triangleq [q^T \ \dot{q}^T]^T \in \mathbb{R}^{20}$, the equations (1) are written as $\dot{x} = \varphi(x, u, d)$,

$$\varphi(x, u, d) = \begin{bmatrix} \dot{q} \\ M^{-1}[-(C + L_{\text{fr}})\dot{q} - g(q) + L_{\text{in}}(q)u + L_{\text{db}}d] \end{bmatrix}, \quad (2)$$

which describes the load behavior in the inertial frame explicitly, with the aircraft's position and orientation being described only with respect to the load.

B. Measurement equation

This work assumes that the load position and orientation with respect to the inertial frame are not measured. Available measurements are provided by: (i) a Global Positioning System (GPS), providing the planar position of the tilt-rotor UAV w.r.t. $\mathcal{I}$; (ii) a barometer, providing the altitude of the UAV w.r.t. $\mathcal{I}$; (iii) an Inertial Measurement Unit (IMU), providing the orientation and angular velocities of the UAV w.r.t. $\mathcal{I}$; (iv) a camera, providing the position of the load with respect to the $\mathcal{A}_1$; and (v) embedded sensors at the servos, providing the tilting angles and their time derivatives.

The GPS, barometer, and IMU are located at $\mathcal{B}$, while the camera is placed at the origin of $\mathcal{A}_1$. Let $\zeta_B \triangleq [x_B \ y_B \ z_B]^T$ denote the position of the UAV w.r.t. $\mathcal{I}$. Then,

$$\zeta_B(\zeta, \eta) = \zeta + R_{\mathcal{L}}^{\mathcal{I}} d_{\mathcal{A}_1}^{\mathcal{L}} + R_{\mathcal{L}}^{\mathcal{B}} R_{\mathcal{B}}^{\mathcal{A}_1} d_{\mathcal{B}}^{\mathcal{A}_1}. \quad (3)$$

¹ $\mathbb{I}_n$ is an identity matrix with dimension n , $\mathbf{0}_{n \times m}$ denotes an n by m matrix of zeros, and $\mathbf{1}_{n \times m}$ is an n by m matrix of ones.

The orientation of the UAV w.r.t. $\mathcal{I}$ is parametrized by $\eta_B \triangleq [\phi_B \ \theta_B \ \psi_B]^T$, using the local roll-pitch-yaw convention. Therefore, $R_B^T \triangleq R_{z,\psi_B} R_{y,\theta_B} R_{x,\phi_B} = R_L^T R_B^C$, hence,

$$\phi_B(\eta, \gamma) = \arctan((R_L^T R_B^C)_{32} / (R_L^T R_B^C)_{33}), \quad (4)$$

$$\theta_B(\eta, \gamma) = \arcsin(-(R_L^T R_B^C)_{31}), \quad (5)$$

$$\psi_B(\eta, \gamma) = \arctan((R_L^T R_B^C)_{21} / (R_L^T R_B^C)_{11}), \quad (6)$$

for $\theta_B \neq \pm\pi/2$, where $(\cdot)_{ij}$ denotes the element from the i -th line and j -th column. Moreover, the angular velocity provided by the IMU is given by

$$\omega_{LB}^B(\eta, \gamma, \dot{\eta}, \dot{\gamma}) = \omega_{LB}^B + \omega_{LB}^C = (R_B^C)^T W_\eta \dot{\eta} + Q \dot{\gamma}. \quad (7)$$

On the other hand, let $d_{A_1 L}^{A_1}$ denote the displacement from A_1 to L , expressed in A_1 , provided by the camera. Then,

$$d_{A_1 L}^{A_1}(\gamma) = -(R_{A_1}^C)^T d_{A_1}^C = -(R_B^C)^T d_{A_1}^C. \quad (8)$$

Putting together (3) through (8), and considering α_R , α_L , $\dot{\alpha}_R$, and $\dot{\alpha}_L$, provided by sensors at the servos, leads to the nonlinear measurement equation $y_k = \pi(x_k) + v_k$, where v_k denotes measurement noise, and

$$\pi(x_k) = \begin{bmatrix} \zeta_B^T & \eta_B^T & (\omega_{LB}^B)^T & (d_{A_1 L}^{A_1})^T & \alpha_R & \alpha_L & \dot{\alpha}_R & \dot{\alpha}_L \end{bmatrix}^T.$$

Finally, let $\mathbb{I}_k$ denote the set of available measurements at time instant k , such that $\mathbb{I}_k \subseteq \{1, 2, \dots, 16\}$, and ι_k denote the number of elements in $\mathbb{I}_k$. Then, $y_k, v_k \in \mathbb{R}^{\iota_k}$, and

$$\pi^{[k]}(x_k) \triangleq [\pi(x_k)(i)]_{i \in \mathbb{I}_k}, \quad (9)$$

where $\pi(x_k)(i)$ denotes the i -th line of $\pi(x_k)$, and the brackets denote vertical concatenation.

III. STATE ESTIMATION BY CONSTRAINED ZONOTOPES

This section describes the set-valued state estimation method used in this work.

A. Set operations

Set operations are often required in set-valued state estimation [4], [9]. Given compact sets $Z, W \subset \mathbb{R}^n$, $Y \subset \mathbb{R}^m$, and a real matrix $R \in \mathbb{R}^{m \times n}$, then the linear image, Minkowski sum, and generalized intersection, are defined as

$$RZ \triangleq \{Rz : z \in Z\}, \quad (10)$$

$$Z \oplus W \triangleq \{z + w : z \in Z, w \in W\}, \quad (11)$$

$$Z \cap_R Y \triangleq \{z \in Z : Rz \in Y\}, \quad (12)$$

respectively. These operations are known to be costly and numerically unstable for convex polytopes in high dimensions [9]. Ellipsoids are closed only under (10). Operations (10) and (11) can be computed efficiently using zonotopes, but (12) is generally not a zonotope [11].

B. Constrained zonotopes

Constrained zonotopes are a generalization of zonotopes, recently proposed in [9]. A compact set $Z \subset \mathbb{R}^n$ is a *constrained zonotope* if there exists $(G, c, A, b) \in \mathbb{R}^{n \times n_g} \times \mathbb{R}^n \times \mathbb{R}^{n_c \times n_g} \times \mathbb{R}^{n_c}$ such that

$$Z = \{c + G\xi : \|\xi\|_\infty \leq 1, A\xi = b\}. \quad (13)$$

As in [9], the shorthand notation $Z = \{G, c, A, b\}$ is used for constrained zonotopes, $Z = \{G, c\}$ is used for zonotopes. Equation (13) is referred to as *constrained generator representation* (CG-rep). Constrained zonotopes correspond

to an *alternative representation* to convex polytopes, and are closed under (10)-(12), which are exactly given by [9]

$$RZ = \{RGz, Rc, Az, bz\}, \quad (14)$$

$$Z \oplus W = \left\{ \begin{bmatrix} G_z & G_w \end{bmatrix}, c_z + c_w, \begin{bmatrix} A_z & 0 \\ 0 & A_w \end{bmatrix}, \begin{bmatrix} b_z \\ b_w \end{bmatrix} \right\}, \quad (15)$$

$$Z \cap_R Y = \left\{ \begin{bmatrix} G_z & 0 \end{bmatrix}, c_z, \begin{bmatrix} A_z & 0 \\ 0 & A_y \\ RG_z & -G_y \end{bmatrix}, \begin{bmatrix} b_z \\ b_y \\ c_y - Rc_z \end{bmatrix} \right\}. \quad (16)$$

Constrained zonotopes are effectively used in [9] for set-valued state estimation of linear systems, and in [12] for fault diagnosis. An important result from [9] follows.

Property 1: For every $Z = \{G, c, A, b\} \subset \mathbb{R}^n$, $Z \neq \emptyset$ iff $\min \{\|\xi\|_\infty : A\xi = b\} \leq 1$, and $z \in Z$ iff $\min_\xi \left\{ \|\xi\|_\infty : \begin{bmatrix} G \\ A \end{bmatrix} \xi = \begin{bmatrix} z - c \\ b \end{bmatrix} \right\} \leq 1$.

This work adapts the estimation method in [9] for discrete-time switching systems, as described in the next subsections.

C. Linearized model for state estimation

Since the method proposed in this work is formulated for linear systems, the state-space equations (2) and measurement mapping (9) are linearized around equilibrium values² x^{eq} , u^{eq} , $d = 0$, through first-order Taylor series expansion, and discretized by the zero-order-hold method, yielding to the disturbed switching linear discrete-time system

$$\Delta x_k = A \Delta x_{k-1} + B \Delta u_{k-1} + F d_{k-1} + w_{k-1},$$

$$y_k = H^{[k]} \Delta x_k + \pi^{[k]}(x^{eq}) + v_k,$$

where the process noise w_k corresponds to unmodeled dynamics due to linearization and discretization, and the measurement noise v_k now includes the nonlinearities of (9). Moreover, $\Delta x_k \triangleq x_k - x^{eq}$ and $\Delta u_k \triangleq u_k - u^{eq}$, with $A \in \mathbb{R}^{20 \times 20}$, $B \in \mathbb{R}^{20 \times 4}$, $F \in \mathbb{R}^{20 \times 3}$, $H^{[k]} \in \mathbb{R}^{\iota_k \times 20}$.

For robust state estimation, the state vector is augmented with the external disturbances d_k , yielding the augmented dynamics

$$\begin{bmatrix} \Delta x_k \\ d_k \end{bmatrix} = \begin{bmatrix} A & F \\ 0_{3 \times 20} & I_3 \end{bmatrix} \begin{bmatrix} \Delta x_{k-1} \\ d_{k-1} \end{bmatrix} + \begin{bmatrix} B \\ 0_{3 \times 4} \end{bmatrix} \Delta u_{k-1} + \begin{bmatrix} w_{k-1} \\ d_{k-1} \end{bmatrix},$$

$$y_k = \begin{bmatrix} H^{[k]} & 0_{\iota_k \times 3} \end{bmatrix} \begin{bmatrix} \Delta x_k \\ d_k \end{bmatrix} + \pi^{[k]}(x^{eq}) + v_k,$$

with $\tilde{d}_{k-1} \triangleq d_k - d_{k-1}$.

Finally, define $\nu_k \triangleq [\Delta x_k^T \ d_k^T]^T$, $\bar{w}_k \triangleq [w_k^T \ \tilde{d}_k^T]^T$, and $\bar{v}_k \triangleq \pi^{[k]}(x^{eq}) + v_k$. Therefore, the set-valued state estimation is formulated in the next section.

D. Problem formulation

Consider the augmented linear discrete-time switching system described in Section III-C, and let A_ν , B_ν and $H_\nu^{[k]}$ denote the associated matrices. Assuming bounded disturbances $\bar{w}_k \in W_k$ and $\bar{v}_k \in V_k$, and $\nu_0 \in X_0$, where X_0 , W_k and V_k are known compact sets (see e.g. the method in [13] for rigorous bounds on the linearization error), set-valued state estimation consists in computing sets $\tilde{X}_k$ and $\tilde{X}_k$ such that

$$\tilde{X}_k \supseteq \{A_\nu \nu + B_\nu \Delta u_{k-1} + \bar{w} : \nu \in \tilde{X}_{k-1}, \bar{w} \in W_{k-1}\}, \quad (17)$$

²Equilibrium values are used in this work since online evaluation of a time-varying model around trajectory values demonstrated intractable.

$$\hat{X}_k \supseteq \{\nu \in \bar{X}_k : H_\nu^{[k]} \nu + \bar{v} = y_k, \bar{v} \in V_k\}, \quad (18)$$

with (17) referred to as *prediction step*, and (18) as *correction step*. Such sets can be obtained exactly with constrained zonotopes, through [9]

$$\bar{X}_k = A_\nu \bar{X}_{k-1} \oplus B_\nu \Delta u_{k-1} \oplus W_{k-1}, \quad (19)$$

$$\hat{X}_k = \bar{X}_k \cap_{H_\nu^{[k]}} (y_k \oplus (-V_k)), \quad (20)$$

using (14)–(16).

E. Minimum-variance feedback

Despite the accuracy and low computational burden of the present method, to perform state-feedback control with estimated states one must choose a point satisfying $\hat{\nu}_k \in \hat{X}_k$. Therefore, this work proposes an optimal choice according to a *minimum-variance criterion*, which is a central contribution of this paper, and can be straightforwardly adapted to more general classes of linear discrete-time switching dynamical systems.

The proposed criterion is based on the well-known Kalman filter algorithm [14]. Let P_k^ν , P_k^w , and P_k^v denote the covariance matrices of the estimation error $\nu_k - \hat{\nu}_k$, process noise $\bar{w}_k$, and measurement noise $\bar{v}_k$, respectively. Consider the constrained optimization problem

$$\min_{N_k} \text{trace}\{P_k^\nu\} \quad \text{subject to} \quad \hat{\nu}_k \in \hat{X}_k, \quad (21)$$

where N_k is the Kalman gain, and $\hat{\nu}_k$ is given according to the correction step

$$\hat{\nu}_k = \bar{\nu}_k + N_k(y_k - (H_\nu^{[k]} \bar{\nu}_k + \pi^{[k]}(x^{\text{eq}}))), \quad (22)$$

with $\bar{\nu}_k = A_\nu \hat{\nu}_{k-1} + B_\nu \Delta u_{k-1}$. Moreover,

$$P_k^\nu = (\mathbb{I}_{23} - N_k H_\nu^{[k]}) \bar{P}_k^\nu (\mathbb{I}_{23} - N_k H_\nu^{[k]})^T + N_k P_k^v N_k^T, \quad (23)$$

where $\bar{P}_k^\nu = A_\nu P_{k-1}^\nu A_\nu^T + P_k^w$. Considering Property 1 and $\hat{X}_k = \{G_x, c_x, A_x, b_x\}$, the optimization problem (21) is rewritten as a quadratic program with linear constraints,

$$\begin{aligned} \min_{N_k, \xi} \text{trace}\{P_k^\nu\} \quad \text{subject to} \quad (24) \\ \bar{\nu}_k + N_k(y_k - (H_\nu^{[k]} \bar{\nu}_k + \pi^{[k]}(x^{\text{eq}}))) = c_x + G_x \xi, \\ \|\xi\|_\infty \leq 1, \quad A_x \xi = b_x, \end{aligned}$$

with P_k^ν given by (23). The minimum-variance choice $\hat{\nu}_k$ is then obtained by solving (24), and computing (22) afterwards. The optimization problem (24) can be regarded as *constrained Kalman filtering*, in which divergence issues are not present since $\hat{\nu}_k \in \hat{X}_k$ is guaranteed by construction.

IV. PATH TRACKING CONTROL

A. Linearized parameter-varying error dynamics

The tilt-rotor UAV with suspended load is an underactuated mechanical system with four control inputs. For path tracking of the suspended load, the position $\zeta = [x \ y \ z]^T$ and yaw angle ψ of the load are chosen to be regulated. Let $q^{\text{tr}} = [\zeta^{\text{tr}} \ \eta^{\text{tr}} \ \gamma^{\text{eq}} \ \alpha_R^{\text{eq}} \ \alpha_L^{\text{eq}}]^T$, with $\eta^{\text{tr}} = [\phi^{\text{eq}} \ \theta^{\text{eq}} \ \psi^{\text{tr}}]^T$, where ‘tr’ denotes trajectory values. Assuming ψ^{tr} constant, the linearized error dynamics of (2) are given by $\delta \dot{x} = A_c(t) \delta x + B_c \delta u + F_c d$, where $\delta x \triangleq x - x^{\text{tr}}$, $\delta u \triangleq u - u^{\text{tr}}$, $x^{\text{tr}} \triangleq [(q^{\text{tr}})^T \ (\dot{q}^{\text{tr}})^T]^T$, and $A_c(t)$ is a function of u^{tr} [8].

Considering a feasible reference trajectory, the desired input can be obtained from (2) as follows:

$$u^{\text{tr}} = L_{\text{in}}(q^{\text{tr}})^+ [M(q^{\text{tr}}) \ddot{q}^{\text{tr}} + (C(q^{\text{tr}}, \dot{q}^{\text{tr}}) + L_{\text{fr}}) \dot{q}^{\text{tr}} + g(q^{\text{tr}})], \quad (25)$$

where $L_{\text{in}}(q^{\text{tr}})^+$ denotes the pseudo-inverse of $L_{\text{in}}(q^{\text{tr}})$. It can be shown that (25) is a function only of the accelerations $\ddot{\zeta}^{\text{tr}}$ [8]. By defining the vector of parameters $\sigma \triangleq \ddot{\zeta}^{\text{tr}}$, we have that $u^{\text{tr}}(t) \triangleq u(\sigma)$, which is affine in σ . Then, substituting (25) in $A_c(t)$ yields the parameter-varying matrix $A_c(\sigma)$, which is affine in σ since (2) is affine in u .

For an improved trajectory tracking and rejection of constant disturbances, the state vector δx is augmented with integral actions over the regulated variables, as $\chi \triangleq [\delta x^T \ \int(\zeta - \zeta^{\text{tr}})^T \ \int(\psi - \psi^{\text{tr}})]^T$. The resulting augmented dynamics are then discretized through Euler approximation, yielding the discrete-time, augmented LPV error dynamics

$$\chi_{k+1} = A_\chi(\sigma) \chi_k + B_\chi \delta u_k + F_\chi d_k, \quad (26)$$

with $A_\chi(\sigma) \in \mathbb{R}^{24 \times 24}$, $B_\chi \in \mathbb{R}^{24 \times 4}$ and $F_\chi \in \mathbb{R}^{24 \times 3}$. Finally, assuming bounded trajectory accelerations, the LPV system (26) can be rewritten in a convex polytopic representation, by defining $A_\chi(\sigma) \triangleq A_\chi(\tilde{\sigma}) = \sum_{i=1}^8 \tilde{\sigma}_i A_\chi^i$, with A_χ^i denoting the i -th vertex of $A_\chi(\tilde{\sigma})$, and $\sum_{i=1}^8 \tilde{\sigma}_i = 1$. Resulting errors from linearization and discretization are taken into account in this section as unmodeled dynamics.

B. Discrete-time mixed $\mathcal{H}_2/\mathcal{H}_\infty$ control with pole placement constraints

The aim of the proposed controller is to perform robust trajectory tracking of the load with stabilization of the tilt-rotor UAV, with estimation provided by the novel method presented in Section III-E. The desired accelerations for the load are required to be smooth and bounded. The present method extends the approach in [15] for discrete-time linear systems. Let

$$\begin{aligned} z_k^{(2)} &= H_z \chi_k + D_{zu} \delta u_k, \\ z_k^{(\infty)} &= H_z \chi_k + D_{zu} \delta u_k + D_{zd} d_k, \end{aligned}$$

where $H_z \in \mathbb{R}^{n_z \times 24}$, $D_{zu} \in \mathbb{R}^{n_z \times 4}$, $D_{zd} \in \mathbb{R}^{n_z \times 3}$. Let $\Psi_{dz}^{(2)}(\varsigma)$ and $\Psi_{dz}^{(\infty)}(\varsigma)$ denote the discrete-time transfer matrices from d_k to $z_k^{(2)}$ and d_k to $z_k^{(\infty)}$, and $\|\Psi_{dz}^{(2)}(\varsigma)\|_2$ and $\|\Psi_{dz}^{(\infty)}(\varsigma)\|_\infty$ be the corresponding $\mathcal{H}_2$ and $\mathcal{H}_\infty$ norms, respectively. Given the state-feedback control law $u_k = -K \chi$, the gain matrix K that minimizes $\text{trace}(\Omega) > \|\Psi_{dz}^{(2)}\|_2^2$ while guaranteeing an upper-bound $\tilde{\gamma} > \|\Psi_{dz}^{(\infty)}\|_\infty^2$, is given by $K = -YX^{-1}$, where Y and X are the solution of the optimization problem

$$\min_{P, X, Y, \Omega} \text{trace}\{\Omega\} \quad \text{subject to} \quad (27)$$

$$\begin{aligned} \begin{bmatrix} \Omega & H_z X + D_{zu} Y \\ * & X + X^T - P \end{bmatrix} &> 0, \\ \begin{bmatrix} P & A_\chi^i X + B_\chi Y & F_\chi & 0_{24 \times n_z} \\ * & X + X^T - P & 0_{24 \times 3} & X^T H_z^T + Y^T D_{zu}^T \\ * & * & \mathbb{I}_3 & D_{zd}^T \\ * & * & * & \tilde{\gamma} \mathbb{I}_{n_z} \end{bmatrix} &> 0, \end{aligned}$$

with $i = 1, 2, \dots, 8$, $P = P^T > 0$, $X > 0$ and $\Omega = \Omega^T$.

To ensure transient properties for the closed-loop system, additional constraints are imposed to pole placement.

Seeking minimum and maximum settling times, maximum percentage overshoot, and also to avoid ringing, three regions are of interest: $\mathbb{D}_1 \triangleq \{\zeta \in \mathbb{C} : \text{Re}(\zeta) > \varepsilon \geq 0\}$, $\mathbb{D}_2 \triangleq \{\zeta \in \mathbb{C} : 0 \leq |\zeta| < \varpi\}$ and $\mathbb{D}_3 \triangleq \{\zeta \in \mathbb{C} : 0 \leq |\text{Im}(\zeta)| < \tau\}$. The eigenvalues of $A_{\chi}(\sigma) - B_{\chi}K$ belong to $\bigcap_{j=1}^3 \mathbb{D}_j$ iff [15]

$$\begin{aligned} X(A_{\chi}^i)^T + A_{\chi}^i X + Y^T B_{\chi}^T + B_{\chi} Y - 2\varepsilon X &> 0, \\ \begin{bmatrix} -\varpi X & A_{\chi}^i X + B_{\chi} Y \\ * & -\varpi X \end{bmatrix} &< 0, \\ \begin{bmatrix} -2\tau X & X(A_{\chi}^i)^T - A_{\chi}^i X + Y^T B_{\chi}^T - B_{\chi} Y \\ * & -2\tau X \end{bmatrix} &< 0. \end{aligned}$$

Finally, the control signals applied to (2) are obtained from $u_k = \delta u_k + u_k^{\text{tr}} = -K\hat{\chi}_k + u_k^{\text{tr}}$, with u_k^{tr} given by (25), where the estimated states $\hat{\chi}_k$ are retrieved from $\hat{\nu}_k$ according to the definitions of the state vectors ν_k and χ_k .

V. NUMERICAL EXPERIMENTS

This section evaluates the performance of the proposed methods through numerical experiments.

The experiment is performed in the ProVANT simulator platform³, using a Computer Aided Design (CAD) model of the system. The model parameters are shown in Table⁴ I (see [10] for definitions), which yields the equilibrium point

$$\begin{aligned} q^{\text{eq}} &= [0_{1 \times 6} \quad 0.000132 \quad 0.01396 \quad 0.01401 \quad 0.01381]^T, \\ u^{\text{eq}} &= [11.7323 \quad 11.7676 \quad 4.1389 \cdot 10^{-7} \quad 1.0121 \cdot 10^{-5}]^T. \end{aligned}$$

TABLE I

MODEL PARAMETERS OF THE SYSTEM.

Parameter	Value
(m_L, m_1, m_2, m_3)	$(0.5, 1.7068, 0.08978, 0.08978)$ Kg
$(d_{A_1}^c, d_{B_1}^{A_1})$	$([0 \ 0 \ 0.5]^T, [0 \ 0 \ 0.119]^T)$ m
$d_{C_1}^B$	$[-0.004321 \ 0.000601 \ -0.045113]^T$ m
$d_{A_2}^{B_1}$	$[0 \ -0.275433 \ 0.056262]^T$ m
$d_{A_3}^{B_2}$	$[0 \ 0.275433 \ 0.056261]^T$ m
$(d_{C_2}^{A_2}, d_{C_3}^{A_3})$	$([0 \ 0 \ 0.05647]^T, [0 \ 0 \ 0.05648]^T)$ m
I_C	$8.333 \cdot 10^{-6} \mathbb{I}_3$ Kg·m ²
I_1	$\begin{bmatrix} 4047.04 & 0.8606 & 9.6577 \\ * & 881.62 & -0.8731 \\ * & * & 4173.18 \end{bmatrix} \cdot 10^{-6}$ Kg·m ²
I_2, I_3	$\text{diag}(335.74, 335.74, 641.59) \cdot 10^{-6}$ Kg·m ²
$\hat{g}$	$[0 \ 0 \ -9.81]^T$ m/s ²
(k_τ, b)	$(1.7 \cdot 10^{-7} \text{ N} \cdot \text{m} \cdot \text{s}^2, 9.5 \cdot 10^{-6} \text{ N} \cdot \text{s}^2)$
$(\lambda_R, \lambda_L, \beta)$	$(1, -1, 5^\circ)$
μ_γ, μ_α	$0.005 \text{ N} \cdot \text{m}/(\text{rad/s})$

The desired trajectory is composed of several paths, with $\psi^{\text{tr}} = 0$ rad. The sensor parameters are shown in Table II.

TABLE II

PARAMETERS OF THE SENSORS.

Sensor	I	Noise bound	Sampling	PDF
GPS	$\{1, 2\}$	± 0.15 m	120 ms	Gauss.
Barometer	$\{3\}$	± 0.51 m	12 ms	Gauss.
IMU	$\{4, 5, 6\}$	$\pm 2.618 \cdot 10^{-3}$ rad	12 ms	Gauss.
	$\{7, 8, 9\}$	$\pm 16.558 \cdot 10^{-3}$ rad/s		
Camera	$\{10, 11\}$	± 0.005 m	24 ms	Unif.
	$\{12\}$	± 0.02 m		
Servos	$\{13, 14\}$	$\pm 5.67 \cdot 10^{-3}$ rad	12 ms	Unif.
	$\{15, 16\}$	± 0.50772 rad/s		

³See <https://github.com/Guiraffo/ProVANT-Simulator>.

⁴Parameter uncertainties can be considered immediately through $\tilde{w}_k, \tilde{v}_k$.

The optimization problem (27) was solved using SDPT3, with $\varepsilon = 0.55$, $\varpi = 0.994$, $\tau = 0.3$, $\hat{\gamma} = 81$, $\hat{x}^{\text{tr}}(t) \in [-0.5, 0.5]$, $\hat{y}^{\text{tr}}(t) \in [-0.5, 0.5]$, $\hat{z}^{\text{tr}}(t) \in [-0.3, 0.3]$,

$$\begin{aligned} H_z &= \text{diag}\left(\frac{\sqrt{10}}{2} \mathbf{1}_{3 \times 1}, \frac{\sqrt{0.5}}{\pi/2} \mathbf{1}_{2 \times 1}, \frac{\sqrt{5}}{\pi}, \frac{2}{\pi} \mathbf{1}_{2 \times 1}, \frac{0.2}{\pi} \mathbf{1}_{2 \times 1}, \right. \\ &\quad \left. \frac{1}{2} \mathbf{1}_{3 \times 1}, \frac{3}{\pi} \mathbf{1}_{2 \times 1}, \frac{4}{\pi}, \frac{\sqrt{5}}{3\pi} \mathbf{1}_{2 \times 1}, \frac{0.1}{3\pi} \mathbf{1}_{2 \times 1}, \sqrt{5} \cdot \mathbf{1}_{3 \times 1}, \sqrt{0.1}\right), \\ D_{zu} &= \begin{bmatrix} \text{diag}\left(\frac{\sqrt{750}}{30-f_R^{\text{eq}}}, \frac{\sqrt{750}}{30-f_L^{\text{eq}}}, \frac{\sqrt{5000}}{2-\tau_{\alpha_R}^{\text{eq}}}\right) & \mathbf{0}_{3 \times 1} \\ \mathbf{0}_{2 \times 3} & \mathbf{0}_{2 \times 1} \\ \mathbf{0}_{1 \times 3} & \frac{\sqrt{5000}}{2-\tau_{\alpha_L}^{\text{eq}}} \\ \mathbf{0}_{18 \times 3} & \mathbf{0}_{18 \times 1} \end{bmatrix}, \\ D_{zd} &= [\mathbf{0}_{3 \times 10} \quad \mathbb{I}_3 \quad V^T \quad 0.5 \cdot \mathbf{1}_{3 \times 1} \quad V^T \quad \mathbf{0}_{3 \times 2} \quad \mathbb{I}_3 \quad 0.5 \cdot \mathbf{1}_{3 \times 1}]^T, \end{aligned}$$

$V \triangleq \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$. The constrained zonotopes X_0 , W_k , and V_k , were chosen as $X_0 = \{G_{x_0}, [(\zeta_0^{\text{tr}})^T \quad \mathbf{0}_{1 \times 20}]^T\}$, $W_k = \{G_w, \mathbf{0}_{23 \times 1}\}$ and $V_k = \{G_v, \pi(x^{\text{eq}})\}$, with $G_{x_0} = \text{diag}(0.5 \cdot \mathbf{1}_{3 \times 1}, 0.02, 0.02, \pi/180, 1.5 \cdot |\gamma_1^{\text{eq}}|, 1.5 \cdot |\gamma_2^{\text{eq}}|, 1.5 \cdot |\alpha_R^{\text{eq}}|, 1.5 \cdot |\alpha_L^{\text{eq}}|, 0.02 \cdot \mathbf{1}_{13 \times 1})$, $G_w = \text{diag}(10^{-4} \cdot \mathbf{1}_{13 \times 1}, 0.01 \cdot \mathbf{1}_{3 \times 1}, 0.05 \cdot \mathbf{1}_{2 \times 1}, 10^{-4} \cdot \mathbf{1}_{2 \times 1}, 0.01 \cdot \mathbf{1}_{3 \times 1})$, $G_v = \text{diag}(0.18 \cdot \mathbf{1}_{2 \times 1}, 0.612, 3.1416 \cdot 10^{-3} \cdot \mathbf{1}_{2 \times 1}, 0.03, 19.872 \cdot 10^{-3} \cdot \mathbf{1}_{2 \times 1}, 0.24, 0.006 \cdot \mathbf{1}_{2 \times 1}, 0.06, 6.8067 \cdot 10^{-3} \cdot \mathbf{1}_{2 \times 1}, 0.6093 \cdot \mathbf{1}_{2 \times 1})$. These were adjusted empirically to accommodate linearization errors. The number of generators and constraints of $\hat{X}_k$ were limited to 50 and 10, respectively, using the complexity reduction methods of [9]. The optimal estimate $\hat{\nu}_k \in \hat{X}_k$ was obtained considering $P_0^v = \mathbb{I}_{23}$, with $\hat{\nu}_0$ as the center of the interval hull⁵ of $\hat{X}_0 = X_0 \cap_{H^v} (y_0 \oplus (-V_0))$, $P_k^w = \frac{1}{3} G_w^2$, and $P_k^v = \frac{1}{3} G_v^2$. The quadratic program (24) was solved using CPLEX. For comparison, estimation results using the zonotope-based algorithm of [8] are also presented, with same X_0 , W_k , and V_k , and number of generators limited to 1725. The initial state is $x_0 = [0 \ 0 \ 1 \ \mathbf{0}_{1 \times 13}]^T$.

The trajectories performed by the load and the UAV are shown in Fig. 2, and the tracking error is shown in Fig. 3. External forces were added and removed from the load, as filtered steps with magnitude 0.5 N. The estimation error of the regulated variables is shown in Fig. 4, along with confidence limits (interval hull), and results using the strategy of [8]. Sharper bounds are observed for the proposed method, which demonstrate the reduced conservatism of constrained zonotopes. This is a direct result of the exact intersection (20), which was severely over-estimated by zonotopic enclosures even with a much higher number of generators. Note that the confidence limits are not symmetric, since the optimal choice ν_k is not the geometric center of $\hat{X}_k$ in general. Moreover, since outliers can happen due to unmodeled dynamics, if $\hat{X}_k = \emptyset$ (verified through Property 1), $\bar{X}_k$ was used instead. The path tracking was performed successfully, as shown by the tracking error of the regulated variables, which remained bounded during the experiment⁶. The control signals applied to the aircraft are shown in Fig. 5. To conclude, experiments were conducted with an alternative feedback $\hat{\nu}_k \in \hat{X}_k$, as the center of the interval hull of

⁵The interval hull is computed as in [9], [16].

⁶A video of the performed experiment is available in <https://youtu.be/x-oJNQqdvds>.

$\hat{X}_k$. Although less computationally demanding [9], such choice demonstrated to be substantially noisy, destabilizing the closed-loop system.

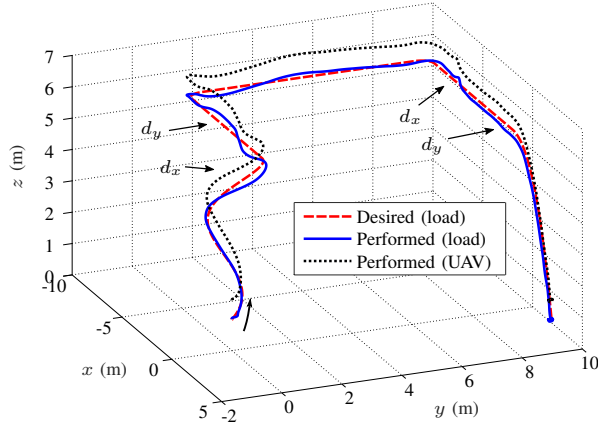


Fig. 2. Trajectories performed by the UAV and the suspended load.

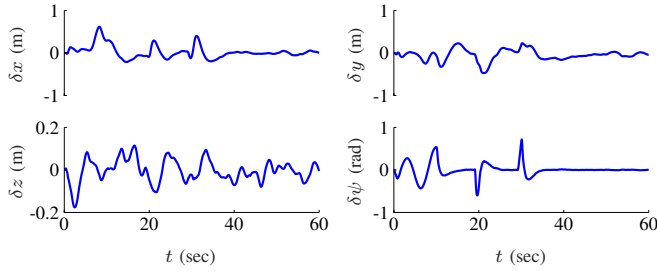


Fig. 3. Tracking error of the regulated variables.

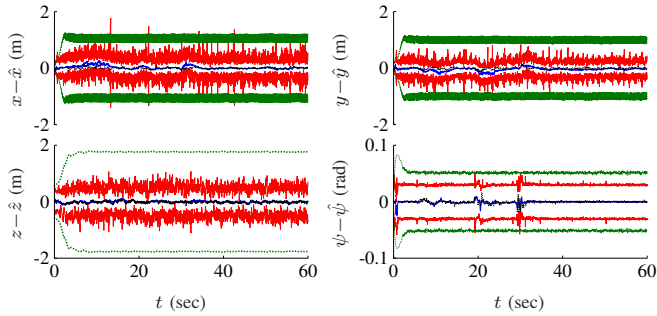


Fig. 4. Estimation error of the regulated variables, along with confidence limits (outer lines). Dotted lines denote results from the algorithm of [8].

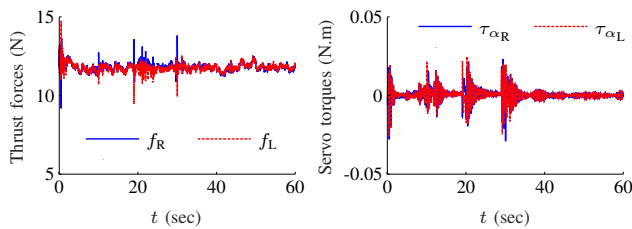


Fig. 5. Control signals applied to the UAV.

VI. CONCLUSION

This paper addressed path tracking of a suspended load by a tilt-rotor UAV, based on set-valued state estimation by constrained zonotopes, to accurately estimate the load position and orientation, considering sensors with different sampling times, and unknown-but-bounded disturbances. Aiming to feedback the controller with a state that belongs to the estimated set, an optimal state choice was proposed based on a constrained minimum-variance criteria, which is the main contribution of this work. Path tracking control was solved by discrete-time mixed H_2/H_∞ control with pole-placement constraints. The proposed method was validated through numerical experiments, demonstrating the reduced conservativeness of the set-valued state estimator based on constrained zonotopes, the performance of the optimal selection of estimated states, and robustness of the designed controller.

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